CT · WEEK 1

CT week 1 patterns: Datasets, variables and iteration

The datasets & datatypes · Variables, iteration & flowcharts. Save this PDF for revision.

THE DATASETS & DATATYPES · IN EVERY PAPER

Know these card layouts

Dataset	One card is	Fields you'll filter on
Scores	One student (30 cards, CardNo 0–29)	Name, Gender (M/F), Date of Birth, Town/City, Mathematics, Physics, Chemistry, Total
Words	One word of a paragraph, in order	Word, Part of Speech, Letter count; a sentence ends at a word with a full stop

THE DATASETS & DATATYPES · IN EVERY PAPER

Know these card layouts

Dataset	One card is	Fields you'll filter on
Shopping Bills	One bill	Shop name, Customer name, items (name, category, quantity, price, cost), bill total
Library, Olympics, others	One book or player record	The question prints the card: read every field name before tracing

THE DATASETS & DATATYPES · IN EVERY PAPER

Key points

- Total = Mathematics + Physics + Chemistry · cost = quantity × price · bill total = sum of item costs · letter count = number of letters in the word.
- Sanity of data: every value must be valid for its type. Marks within 0–100, Gender ∈ {M, F}, real dates, totals that add up. A card that breaks these is an error.

THE DATASETS & DATATYPES · IN EVERY PAPER

Datatypes

Type	Values	Operations that make sense
Boolean	True, False	and, or, not
Integer	…, –1, 0, 1, …	+ − × ÷ mod, comparisons
Character	One letter or symbol	==, !=, alphabetical order
String	A sequence of characters (names, words)	==, joining, length
Subtypes	Marks (integer 0–100), Gender (M/F), Date	Only what the meaning allows: adding two names is invalid

Flowchart symbols

Shape	Meaning
Oval / rounded rectangle	Start or End (terminal)
Rectangle	Process, e.g. count = count + 1
Diamond	Decision: a condition with Yes and No exits
Parallelogram	Input or output
Arrow	Flow of control

VARIABLES, ITERATION & FLOWCHARTS · OFTEN ASKED

Key points

- Iterator: Pile 1 (unseen) → pick card X → process X → move X to Pile 2 → repeat until Pile 1 is empty.
- Initialise every variable before the loop (count = 0, sum = 0).
- Filtering: process a card only when a condition holds (a decision box, or an if).
- Average = sum ÷ count, computed after the loop ends.

VARIABLES, ITERATION & FLOWCHARTS · OFTEN ASKED

Count + sum in one pass

```
count = 0
sum = 0
while(Table 1 has more rows){
    Read the first row X in Table 1
    if(X.Gender == "F"){
        count = count + 1
        sum = sum + X.Mathematics
    }
    Move X to Table 2
}
avg = sum / count
```

avg = average Maths marks of female students.

VARIABLES, ITERATION & FLOWCHARTS · OFTEN ASKED

Common traps

⚠ Initialising inside the loop resets the variable on every card, so it ends up holding only the last card's value.

⚠ A loop that never moves X to Table 2 never ends.

Notes for every week of CT, free.

Full notes, the pattern sheet and common traps, with worked PYQs.

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